

Element K

Summary:

The sensory room at Whitmarsh Elementary did not provide enough stimulation to the kids in the class. The group started off researching functions of a sensory room. The main objective of a sensory room is an area where kids can go to relax and decompress. After the basic research the group analyzed specific sensory devices to inspire a design to create an original unique design. Fiber optics caught the group's attention because of their light patterns and flexibility in the strands. Using the flexibility of the fiber optics it allowed the group to arrange them into a wave pattern. After identifying the constraints based on the stakeholders needs. Next, designing the design in Onshape and getting the design approved. Then the group moved onto building the design. The building plans stayed the same throughout the project. Beside one issue the group ran into where we had to adjust the amount of strands in each groove. Lastly, once building was complete the group moved onto testing the project based on the constraints. Overall the group is happy with the product produced.



Successes:

Most of the success the group experienced came in the preliminary planning phases of the project. Prior to construction, the team researched multiple similar project ideas and sources of inspiration, researched different methods of visual stimulations, and conferred with project stakeholders to establish a project design suited to the convenience of the classroom students and teachers. The group communicated

strongly and frequently with the stakeholders and relevant experts throughout the preliminary planning process. Ultimately, the group found success in the final pleasing appearance of the Fiber Optic Waveboard. Despite setbacks in construction, the final project is well constructed and visually appealing.

Drawbacks:

One major drawback the group experienced was drilling the acrylic into the board. On the first attempt we pre-drilled a hole smaller than the screw that caused it to crack the corner. On the next attempt it cracked again, but this time it happened when the hole was being pre-drilled. As a result we pre drilled with a drill bit smaller than the screw. Then pre-drilled again with a drill bit

bigger than the screw and the group was successful. Another drawback the group experienced was staining the wood. No one in the group had prior experience with staining, because of this it took three times for the stain to reach the group's standard of success. Also the grooves where the fiber optics are glued are not the same width throughout. We planned to put seven strands of fiber optics into the grooves but the group was only able to fit five strands.

Lessons Learned:

The group learned many valuable lessons throughout the project. One lesson we learned is to research the type of material that is going to be used in the project. The group did not confirm that it is necessary to predrill a hole bigger than the screw and it delayed the project. Another valuable lesson the group learned was how to apply stain on wood. The group started out applying the stain in a circular motion. Then after sanding the board and restarting the group learned it was better to start in a circular motion and then wipe across horizontally to get rid of the excess stain.